

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

تقييم فعالية مترابك الفضة/أكسيد الزنك النانوي المعلق في بيفا371 في الحفاظ على المنسوجات التراثية: دراسة تجريبية وتطبيقية

Nabil Mabrouk

Faculty of Archaeology - Damietta University

nsh00@du.edu.eg

Abstract

This study focused at how well Beva371 copolymer mixed with Ag/ZnO Nanocomposite (Ag/ZnO NC) works to preserve a 15th-16th century AD Turkish tablecloth made of plain cotton base embroidered with silk threads, which was dyed with natural dyes of indigo, cochineal, weld, and madder. Blank cotton and indigo-dyed silk fabric mock-ups were prepared similarly to the case-study object and treated with different concentrations of the prepared Ag/ZnO NC, then assessed. The implemented characterization techniques confirmed the successful preparation of Ag/ZnO NC, and the suitability of the 800 µg/mL Ag/ZnO NC suspended in 2% Beva371 in the treatment of mock-ups. The obtained results confirmed the improvement in the properties of treated mock-ups without notable side effects, especially in stiffness, color, and air permeability. After accelerated ageing, the treated mock-ups exhibited high resistance against UV, heat, moisture, fungi (especially *A. niger*), and bacteria (especially *E. coli*). After characterization, the case-study object was then cleaned, deacidified, consolidated with the Beva371-Ag/ZnO NC, and supported with needlework on a new blank linen fabric stretched on a treated wooden frame.

Keywords: Ag/ZnO Nanocomposite, Textile Conservation, Consolidation, Anti-UV, antimicrobial.

المخلص:

ركزت هذه الدراسة على مدى فعالية كوبوليمر Beva371 مع مترابك Ag/ZnO (Ag/ZnO-NC) في الحفاظ على مفرش طاولة تركي يرجع إلى القرن 15-16م مصنوع من أرضية من قطن سادة مطرزة بخيوط الحرير، المصبوغة بصبغات طبيعية من النيل، والكوشنيل، والوسمة، والفوة. تم تحضير نماذج حديثة من قماش القطن السادة والحرير المصبوغ بالنيلة الطبيعية بطريقة مشابهة للأثر موضوع الدراسة، وتمت معالجتها بتركيزات مختلفة من Ag/ZnO NC المحضر. أكدت تقنيات الدراسة نجاح تحضير Ag/ZnO NC، وملائمة بتركيز 800 ميكروغرام/مل من Ag/ZnO NC معلقاً في بيفا371 تركيز 2% في معالجة النماذج التجريبية. أكدت النتائج التي تم الحصول عليها تحسن خواص النماذج المعالجة دون آثار جانبية ملحوظة، خاصة في الصلابة واللون ونفاذية الهواء. بعد التقدم المعجل، أظهرت النماذج التجريبية المعالجة مقاومة عالية ضد الأشعة فوق البنفسجية، والحرارة، والرطوبة، والفطريات (خاصة *A. niger*)، والبكتيريا (خاصة *E. coli*). وبعد الفحص والتحليل، تم تنظيف، ومعادلة حموضة، وتقوية الأثر موضوع الدراسة باستخدام Beva371-Ag/ZnO NC، وتدعيمه بشغل الإبرة بواسطة قماش كتان جديد مشدود على إطار خشبي معالج.

الكلمات الدالة: مترابك Ag/ZnO النانوي، صيانة النسيج، تقوية، مضاد لأشعة UV، مضاد للميكروبات.

1. Introduction

Heritage objects are those that have great historical, social, cultural, and educational value. Preservation of such heritage objects is of the utmost importance for future generations. Heritage objects are subject to three major deterioration factors: physical, chemical and biological agents, mainly the fluctuations in temperature and relative humidity, microbial growth¹. Most heritage objects are made up of different materials of organic and inorganic origin, which have different coefficients to different deterioration factors, resulting in a decrease in their elasticity, strength, and cohesion^{2,3,4}. Fluctuating variations in temperature and relative humidity can also be detrimental to heritage materials.

Heritage objects can absorb and then evaporate water molecules; consequently, these objects will expand and contract in response to fluctuations in temperature and relative humidity, resulting in different deterioration phenomena such as brittleness, weakness, flaking, structural bending, color changes, microbial attack, etc.^{5,6,7}. The higher the relative humidity, the higher the potential for microbial growth. Lower temperatures in confined spaces increase the relative humidity, resulting in surface condensation. Moreover, the lower the relative humidity, the higher the loss of textile flexibility^{8,9}. Microorganisms (mainly fungi and bacteria) can affect heritage textile

¹ Guo, C., Lan, L., Liu, Y., Meng, N. and Li, C., 2023. Comparison of environmental criteria for conservation and storage of collections: A comprehensive literature review. *Building and Environment*, 243, p.110665.

² Ilies, D.C., Zlatev, Z., Ilies, A., Zharas, B., Pantea, E., Hodor, N., Indrie, L., Turza, A., Taghiyari, H.R., Caciara, T. and Costea, M., 2022. Interdisciplinary research to advance digital imagery and natural compounds for eco-cleaning and for preserving textile cultural heritage. *Sensors*, 22(12), p.4442.

³ Samadelli, M., Zink, A.R., Roselli, G., Gabrielli, S., Tabandeh, S. and Fernicola, V.C., 2019. Development of passive controlled atmosphere display cases for the conservation of cultural assets. *Journal of Cultural Heritage*, 35, pp.145-153.

⁴ Guo, C., Lan, L., Liu, Y., Meng, N. and Li, C., 2023. Comparison of environmental criteria for conservation and storage of collections: A comprehensive literature review. *Building and Environment*, 243, p.110665.

⁵ Bratasz, Ł., Łukomski, M., Klisińska-Kopacz, A., Zawadzki, W., Dzierżęga, K., Bartosik, M., Sobczyk, J., Lennard, F.J. and Kozłowski, R., 2015. Risk of Climate-Induced Damage in Historic Textiles. *Strain*, 51(1), pp.78-88.

⁶ Jakiela, S., Bratasz, Ł. and Kozłowski, R., 2008. Numerical modelling of moisture movement and related stress field in lime wood subjected to changing climate conditions. *Wood Science and Technology*, 42, pp.21-37.

⁷ Michalski, S., 2007. The ideal climate, risk management, the ASHRAE chapter, proofed fluctuations, and towards a full risk analysis model. *Experts roundtable on sustainable climate management strategies*, pp.1-19.

⁸ Guo, C., Lan, L., Liu, Y., Meng, N. and Li, C., 2023. Comparison of environmental criteria for conservation and storage of collections: A comprehensive literature review. *Building and Environment*, 243, p.110665.

⁹ Mackawa, S., Beltran, V.L. and Henry, M.C., 2015. *Environmental Management for Collections: Alternative Conservation Strategies for Hot and Humid Climates* (). Getty Publications.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

materials by either using fibers as a nutrient source, and/or secreting metabolites¹⁰. The prevalent species of fungi isolated from heritage textiles were *Aspergillus* spp., *Cladosporium* spp., *Fusarium*, *Epiconum* spp. *Paecilomyces* spp., *Penicillium* spp., *Rhizopus* spp., *Streptomyces* spp., *Trichoderma* spp., etc., while the prevalent species of bacteria colonizing heritage textiles were *Arthrobacter* spp., *Bacillus* spp., *Georgenia* spp., *Staphylococcus* spp., *Ornithinibacillus* spp., etc.^{11,12,13,14,15}.

To gather the friable fibers and impart physical strength to the textile fabric, a consolidation process is the aimed option. To achieve that, a substance in a liquid phase is usually introduced into the textile structure and then solidified. Many materials were used as consolidants in the past, such as cellulose nitrate, paraffin, beeswax, shellac, starches, casein, gelatin, etc.¹⁶. The addition of some nanomaterials to these polymers will modify the properties of the treated textiles, and give them new and better properties, such as friction resistance, antimicrobial resistance, water-repellent properties, and protection from ultraviolet radiation^{17,18,19}.

ZnO nanoparticles (ZnO-NPs) attracted great research attention because of many distinct properties such as high stability, easy fabrication, easy application, effectiveness

¹⁰Pekhtasheva, E.L., Neverov, A.N., Kubica, S. and Zaikov, G.E., 2011. Biodegradation and biodeterioration of some natural polymers. *Polymers Research Journal*, 5(1), p.77.

¹¹ Abo-Elmaaref, M., Marouf, M., Mohamed, W.S. and Abdel Wahab, W.A., 2023. Antifungal and consolidation properties of linen textiles treated with silver nanoparticles loaded on hydroxypropyl cellulose polymer. *Heritage Science*, 11(1), p.120.

¹² Ilieș, D.C., Hodor, N., Indrie, L., Dejeu, P., Ilieș, A., Albu, A., Caciara, T., Ilieș, M., Barbu-Tudoran, L. and Grama, V., 2021. Investigations of the surface of heritage objects and green bioremediation: case study of artefacts from Maramureș, Romania. *Applied Sciences*, 11(14), p.6643.

¹³ Kakakhel, M.A., Wu, F., Gu, J.D., Feng, H., Shah, K. and Wang, W., 2019. Controlling biodeterioration of cultural heritage objects with biocides: A review. *International Biodeterioration & Biodegradation*, 143, p.104721.

¹⁴ Pietrzak, K., Puchalski, M., Otlewska, A., Wrzosek, H., Guiamet, P., Piotrowska, M. and Gutarowska, B., 2017. Microbial diversity of pre-Columbian archaeological textiles and the effect of silver nanoparticles misting disinfection. *Journal of Cultural Heritage*, 23, pp.138-147.

¹⁵ Wendt, J.A., Indrie, L., Dejeu, P., Albu, A., Ilieș, D.C., Costea, M., Caciara, T., Ilieș, G., Hodor, N., Josan, I. and Berdenov, Z., 2021. Natural sources in preventive conservation of naturally aged textiles.

¹⁶ Tímár-Balázsy, Á. and Eastop, D., 2012. *Chemical principles of textile conservation*. Routledge.

¹⁷ Al-Gaoudi, H.A., Marouf, M.A., Badry, N. and Rehan, M., 2024. A facile approach to developing multifunctional archaeological wool fabric surface with self-cleaning, and UV protection properties using nano-coating. *Inorganic Chemistry Communications*, 167, p.112716.

¹⁸ Mandal, S., Kumar, P., Satpathy, B., Das, K. and Das, S., 2024. Nanostructured metal oxide based coating for the protection and conservation of cultural heritage: A comprehensive review. *Journal of Cultural Heritage*, 69, pp.94-112.

¹⁹ Spielman-Sun, E., Zaikova, T., Dankovich, T., Yun, J., Ryan, M., Hutchison, J.E. and Lowry, G.V., 2018. Effect of silver concentration and chemical transformations on release and antibacterial efficacy in silver-containing textiles. *NanoImpact*, 11, pp.51-57.

at low concentrations, and high efficiency against microbes, UV-blocking, photocatalytic, and self-cleaning applications^{20,21,22}. In addition to its capability to destroy the growth of microbes²³, ZnO-NPs act as an effective bactericidal cell toxicity and are non-toxic to human cells. Moreover, ZnO-NPs are very efficient in the neutral pH environments in the absence of light. Also, ZnO-NPs have chemical stability under UV radiation and high temperature²⁴. Previous studies revealed that ZnO-NPs aid in improving the properties of the polymers, therefore, it has been combined with some polymers to consolidate many heritage organic and inorganic materials^{25,26}. Ag-NPs are also used as antimicrobial agents against the biodeterioration affecting textile objects. Ag-NPs have been used, as well as ZnO-NPs, as antibacterial against *E. coli*, *B. subtilis*, *P. aeruginosa*, and *K. pneumoniae*^{27,28}. For more efficacy, Ag-NPs were added to other nanoparticles, such as ZnO-NPs and TiO₂-NPs, to provide a double antimicrobial impact. Similarly, Co-NPs and F-NPs were added to ZnO-NPs to study their antimicrobial effect^{29,30}.

The present study focuses on an experimental assessment of Ag-NPs loaded on ZnO-NPs to form Ag/ZnO nanocomposite (Ag/ZnO-NC), due to its efficacy as an antimicrobial, anti-UV, and water-repellent nanocomposite, dispersed in Beva371

²⁰ Dhage, S. R., Renu Pasricha, and V. Ravi. "Synthesis of fine particles of ZnO at 100 C." *Materials Letters* 59, no. 7 (2005): 779-781.

²¹ Tiwari, V., Mishra, N., Gadani, K., Solanki, P.S., Shah, N.A. and Tiwari, M., 2018. Mechanism of anti-bacterial activity of zinc oxide nanoparticle against carbapenem-resistant *Acinetobacter baumannii*. *Frontiers in microbiology*, 9, p.1218.

²² Becerra, J., Ortiz, P., Zaderenko, A.P. and Karapanagiotis, I., 2020. Assessment of nanoparticles/nanocomposites to inhibit micro-algal fouling on limestone façades. *Building Research & Information*, 48(2), pp.180-190.

²³ Varaprasad, K., Raghavendra, G.M., Jayaramudu, T. and Seo, J., 2016. Nano zinc oxide–sodium alginate antibacterial cellulose fibres. *Carbohydrate polymers*, 135, pp.349-355.

²⁴ Trandafilović, L.V., Božanić, D.K., Dimitrijević-Branković, S., Luyt, A.S. and Djoković, V., 2012. Fabrication and antibacterial properties of ZnO–alginate nanocomposites. *Carbohydrate Polymers*, 88(1), pp.263-269.

²⁵ Ibrahim, M.M., Mohamed, W.S. and Mohamed, H.M., 2021. Experimental study for evaluation of Paraloid® B72 and its nanocomposite with nano TiO₂ and nano ZnO for consolidation of pottery samples. *Scientific culture*, 7(2), pp.101-111.

²⁶ Ion, R.M. and Grigorescu, R.M., 2018. Polymeric Micro-and Nanosystems for Wood Artifacts. *New Uses of Micro and Nanomaterials*, p.73.

²⁷ Ruparelia, J.P., Chatterjee, A.K., Duttagupta, S.P. and Mukherji, S., 2008. Strain specificity in antimicrobial activity of silver and copper nanoparticles. *Acta biomaterialia*, 4(3), pp.707-716.

²⁸ Perkas, N., G. Amirian, S. Dubinsky, S. Gazit, and A. Gedanken. "Ultrasound-assisted coating of nylon 6, 6 with silver nanoparticles and its antibacterial activity." *Journal of applied polymer science* 104, no. 3 (2007): 1423-1430.

²⁹ Podporska-Carroll, J., Myles, A., Quilty, B., McCormack, D.E., Fagan, R., Hinder, S.J., Dionysiou, D.D. and Pillai, S.C., 2017. Antibacterial properties of F-doped ZnO visible light photocatalyst. *Journal of hazardous materials*, 324, pp.39-47.

³⁰ Oves, M., Arshad, M., Khan, M.S., Ahmed, A.S., Azam, A. and Ismail, I.M., 2015. Anti-microbial activity of cobalt doped zinc oxide nanoparticles: Targeting water borne bacteria. *Journal of Saudi Chemical Society*, 19(5), pp.581-588.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

acrylic copolymer, due to its remarkable and significant role in the consolidation of heritage textiles. This formula was tested on blank cotton and indigo-dyed silk fabric mock-ups, generally similar to components of the case-study object. The optimum results obtained from the experiments were applied to the case-study heritage tablecloth displayed in the Agricultural Museum in Egypt, after documentation, identification of its components, identification of the deteriorative microorganisms, and treatment. Loading Ag-NPs on ZnO-NPs and mixing it with Beva371 appeared a very promising approach and is introduced for the first time in the conservation of heritage textiles. It greatly improved the properties of the treated fabric in their context without notable side effects.

2. Materials and Methods

2.1. Case-study object

The case-study object (Fig. 1) is a heritage Turkish tablecloth displayed in the Agricultural Museum in Egypt under museum no. 174 in the Heritage Collection building. The tablecloth is a 112 x 51 cm size, dating back to the Ottoman Period (15th-16th century AD) in Turkey. It consists of a blank cotton base embroidered with silk threads dyed with different colors. The decoration consists of a wide band forming a rectangular shape of blue ornaments on a blank background. It is based on 12 groups of Chinese clouds, Chi Chi, which were derived from Chinese traditions, in addition to the Ottoman Turkish decorations, which consist of leaves, branches, and pomegranate buds. These decorations closely resemble the botanical elements found on Turkish ceramics and tiles in the 15th and 16th centuries AD.



Fig. 1 The case-study Turkish tablecloth, (a) recto, (b) verso, (c) deterioration phenomena.

The object suffers many deterioration phenomena (Fig. 1c), like brittleness, weakness, high acidity, dust and stains, tears and ruptures, biodeterioration, etc. Eight microsamples were collected to identify the fibers, dyes, and mordants. Swab samples were collected to identify biodeterioration agents.

2.2. Characterization techniques

2.2.1. Stereo microscope (SM):

A stereo microscope (Olympus CX31, Japan) was used to identify the types of microbes isolated from the case-study object.

2.2.2. Scanning electron microscope (SEM):

A scanning electron microscope (JEOL JSM-6510LB, Tokyo, Japan) coupled with the EDAX was used to identify the types of fibers and the deterioration forms in the case-study object. It was also used to characterize the hereafter-mentioned prepared Ag/ZnO-NC. EDAX unit was used to identify the inorganic materials in the case-study object, especially mordants, and to characterize the prepared Ag/ZnO-NC as well³¹. The samples were mounted on aluminum stubs using carbon discs. The beam energy was 20 kV, the acceleration voltage was at 10 mm working distance and a spot size of 5.5, the spot size was 7 with a scale of approximately 200µm.

2.2.3. Transmission electron microscopy (TEM) and X-ray diffraction (XRD)

Transmission electron microscope (JEOL, JEM 2100F, Tokyo, Japan) was used to characterize the size and morphology of the hereafter-mentioned Ag/ZnO-NC at 200 kV. A drop of a colloidal solution was added to a 400-mesh copper grid coated with carbon film, then the solvent was evaporated at room temperature³². XRD (Bruker D8 ADVANCE, Karlsruhe, Germany) was used to characterize the crystalline nature of the prepared Ag/ZnO-NC³³.

2.2.4. MALDI-TOF-MS

Matrix-assisted Laser Desorption/ionization Time-of-Flight Mass Spectrometry (MALDI-TOF-MS) (Autoflex Speed Bruker, Germany) equipped with 337 nm nitrogen laser in negative reflector mode was used to identify the dyes used in coloring the threads of embroidery in the case-study. The mass spectra for each microfiber were taken between 0 and 5000 *m/z*. The collected laser shots for each spectrum were

³¹Mtavangu, S.G., Machunda, R.L., van der Bruggen, B. and Njau, K.N., 2022. In situ facile green synthesis of Ag–ZnO nanocomposites using Tetradeniariperia leaf extract and its antimicrobial efficacy on water disinfection. *Scientific Reports*, 12(1), p.15359.

³²Elsayed, Y. and Sayed-Ahmed, K., 2024. A Novel Approach of Using Green Synthesized Tellurium Nanoparticles to Protect Historical Oil Paintings from Bacterial Degradation. *Shedet*, 13(13).

³³Noohpisheh, Z., Amiri, H., Farhadi, S. and Mohammadi-Gholami, A., 2020. Green synthesis of Ag-ZnO nanocomposites using Trigonella foenum-graecum leaf extract and their antibacterial, antifungal, antioxidant and photocatalytic properties. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 240, p.118595.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

processed with Bruker Flex III, Bruker Xtof software and mMass 5.0.1 software (Bruker)^{34,35}.

2.2.5. Color measurements

The surface colors of the hereafter-mentioned prepared fabric mock-ups were measured using a colorimeter (PCE-CSM 2, PCE, Germany). Illuminant D65, SCI mode, 10° supplementary standard observer. L*, a*, b* were determined. (ΔE^*) were calculated according to $\Delta E = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}$ equation³⁶. Color measurements of the prepared fabric mock-ups aimed at clarifying the possible color effects of the studied treatments on the fabric properties.

2.2.6. Tensile strength (TS) and elongation (E%)

To clarify how far the studied treatments improved the mechanical properties of the tested fabric mock-ups after treatment, a universal testing machine UTM (Testometric XFS 150 KN, UK) was used to assess the tensile strength (TS) and elongation (E%) of the tested mock-ups before and after treatment according to ASTM (2000), D 5035-95. The test speed was 25 mm/min., and the spacing of the initial jaw was 50 mm³⁷.

2.2.7. Stiffness Test (BL)

To clarify how far the studied treatments increased the stiffness of the fabric mock-ups, a stiffness tester (Full-Automatic Stiffness Tester GT-C70A) was used to measure the bending length (BL) according to ASTM D 1388³⁸. The sample size was 25×250 mm, the press plate speed was 0.4 cm/s, the angle of bevel was 45°.

2.2.8. Air permeability and weight tests

To clarify how far the studied approaches decreased/increased the air permeability of fabric samples, an automatic air permeability tester (GT-C27) was used. The test has been implemented at a pressure of 100 Pa.cm³/cm²/s in accordance with ISO 9237³⁹. To assess the weight of fabric samples and to clarify how far the studied treatments increased the weight of the mock-ups, a 4-decimal lab balance (OHAUS PR224, USA) was used.

³⁴ Mabrouk, N., 2020. Archaeometrical study of a rare embroidered and applique leather tapestry from the safavid artworks. part i: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, 20(1), pp.163-163.

³⁵ Romero-Pastor, J., Navas, N., Rodriguez-Simon, L., Lario-Simón, A., Kuckova, S. and Manzano, E., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry to the Study of the Proteinaceous Binders in Paint, "The Life of Virgin" by Alonso Cano (17th Century) as a Case Study. *European Journal of Mass Spectrometry*, (2015), 21(1), pp.19-26.

³⁶ Marchiafava, V., Bartolozzi, G., Cucci, C., De Vita, M. and Picollo, M., 2014. Colour measurements for monitoring the conservation of contemporary artworks. *Journal of the International Colour Association*, 13, pp.36-42.

³⁷ ASTM International, 2004. *Annual book of ASTM standards*. ASTM international.

³⁸ Sengupta, S., Debnath, S. and Sengupta, A., 2016. Fabric bending behaviour testing instrument for technical textiles. *Measurement*, 87, pp.205-215.

³⁹ Ogulata, R.T., 2006. Air permeability of woven fabrics. *Journal of Textile and Apparel, Technology and management*, 5(2), pp.1-10.

2.3. Chemicals and raw materials

Scoured blank cotton and silk fabrics from the local market were used to prepare the mock-ups to make case-study simulated modern samples. The cotton weaving structure is plain 1/1, the number of threads is 43 warps and 51 wefts per cm, weight is 235 gm/m². The silk weaving structure is plain 1/1, the number of threads is 46 warps and 50 wefts per cm, weight is 75 gm/m². Natural indigo (Wild colours, UK) was used to dye silk fabrics in blue color. Beva371(CTS, Italy) was used as consolidating material, dissolved in toluene (Sigma-Aldrich), and reinforced with the prepared nanocomposite as shown hereafter. Zinc oxide (ZnO), silver nitrate (AgNO₃), ascorbic acid (C₆H₈O₆), sodium hydroxide (NaOH), and sodium hydrosulfite (Na₂S₂O₄.H₂O) (Sigma-Aldrich) were used to prepare the nanocomposite “without further purification” and dye the silk fabrics with indigo. Potato dextrose agar (PDA) and nutrient agar (NA) (Sigma-Aldrich) were used as media for fungal growth and bacterial growth, respectively.

2.4. Identification of microbial strains

Fungal and bacterial strains were isolated from the case-study collected swabs on PDA and NA plates, respectively⁴⁰. For the purification of the isolated fungi, the single spore or the hyphal tip techniques were used. For the purification of the isolated bacteria, the single colony technique was used. The isolated fungi and bacteria were transferred into PDA and NA slants, respectively and kept at 4°C^{41,42}. The isolated fungi and bacteria were identified according to the microscopic characteristics and simple staining, Gram and spore staining^{43,44,45}. The obtained results identified 1) fungi: *Aspergillus niger* (*A. niger*), *Fusarium solani* (*F. solani*), *Aspergillus flavus* (*A. flavus*), and *Cladosporium halotolerans*, (*C. halotolerans*); 2) gram-positive bacteria: *Staphylococcus aureus* (*S. aureus*), *Streptomyces albus* (*S. albus*), and *Bacillus cereus* (*B. cereus*); 3):Gram-negative bacteria: *Escherichia coli* (*E. coli*), *Pseudomonas aeruginosa* (*P. aeruginosa*), *Candida albicans* (*C. albicans*), and *Salmonella typhi* (*S. typhi*).

2.5. Preparation of fabric mock-ups

The blank cotton and silk fabrics were immersed and stirred at 60°C for 30 min. in hot water mixed with commercial soap, then rinsed and dried in an oven at 50°C (ASTM- D629, 1999). Silk fabric was dyed with indigo dye according to the common

⁴⁰ Akhter, M.S., Hossain, S.J., Hossain, S.A. and Datta, R.K., 2012. Isolation and characterization of salinity tolerant *Azotobacter* sp. *Greener J Biol Sci*, 2(3), pp.43-51.

⁴¹ Bhaduri, J., Kundu, P., Mitra, D. and Roy, S.K., 2016. Isolation and characterisation of Nitrogen Fixing Bacteria (*Azotobacter* sp.) from Tea field soil of Dooars and Darjeeling region of North Bengal, India. *Int. J. Eng. Sci. Invent*, 5, pp.46-51.

⁴² Domsch, K.H., Gams, W. and Anderson, T.H., 1980. Compendium of soil fungi. Volume 1.

⁴³ Summerell, B.A., Salleh, B. and Leslie, J.F., 2003. A utilitarian approach to *Fusarium* identification. *Plant disease*, 87(2), pp.117-128.

⁴⁴ Domsch, K.H., Gams, W. and Anderson, T.H., 1980. Compendium of soil fungi. Volume 1.

⁴⁵ Akhter, M.S., Hossain, S.J., Hossain, S.A. and Datta, R.K., 2012. Isolation and characterization of salinity tolerant *Azotobacter* sp. *Greener J Biol Sci*, 2(3), pp.43-51.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

recipes^{46,47}. Mock-ups (Ms) were prepared from cotton fabric (CMs) and silk fabric (SMs) according to the required size for each test. All Ms were then aged at 125 °C for 21 days at oven (Binder ED115, Germany), to create fiber properties like the naturally aged cotton and silk heritage textiles^{48,49}.

2.6. Preparation of Ag/ZnO-NC

0.5 g of ZnO-NPs was mixed with 0.5 g of AgNO₃ and milled manually for 30 min. in an agate mortar. The mixture was then kept for 24 hr. to achieve adsorption of the Ag precursor on the surface of the ZnO particles. Then, 0.518 g of ascorbic acid was added to the mixture and ground for 30 min. The obtained powder containing Ag-NPs was stored in a refrigerator at 4 °C until further use.

2.7. Preparation of acrylic-based nanocomposites (ABNC)

The Ag/ZnO-NC was suspended in a solution of 2% Beva371 acrylic copolymer in toluene in different concentrations (400, 600, 800, 1000 µg/mL). The four concentrations will be named as 1, 2, 3, and 4 respectively in the hereafter characterization. After suspension, the solutions were sonicated (Elmasonic S S30H, Germany) at 22° C for 30 min. to ensure the complete distribution of the Ag/ZnO-NC in Beva371 solution⁵⁰.

2.8. Treating and ageing of mock-ups

After sonication of Ag/ZnO-NC in Beva solutions, the CMs and SMs were treated by immersing in the four concentrations of ABNC at 22° C for 1 min. under sonication. Samples treated with the ABNC (T-CMs, T-SMs) were then dried and kept for 72 hrs. at 22° C⁵¹. The dried T-CMs and T-SMs were then exposed to accelerated ageing (TA-CMs, TA-SMs) to assess the efficiency of the tested treatments in fiber consolidation, and anti-UV activities for the long term. Samples were exposed to the UV radiation in the UV chamber (QUV-se, USA) for one month. Each cycle consists of two steps: 1) the UV weathering at 60°C, UVA 340 nm lamps, 1.5 W/m²/nm intensity, 2) the

⁴⁶ Saez, N.O., Vanden Berghe, I., Schalm, O., De Munck, B. and Caen, J., 2019. Material analysis versus historical dye recipes: ingredients found in black dyed wool from five Belgian archives (1650-1850). *ConservarPatrimónio*, 31, pp.115-132..

⁴⁷ Bechtold, T., Turcanu, A., Ganglberger, E. and Geissler, S., 2003. Natural dyes in modern textile dyehouses—how to combine experiences of two centuries to meet the demands of the future?. *Journal of Cleaner Production*, 11(5), pp.499-509.

⁴⁸ Abdel-kareem, O., 2015. Preparation of experimental deteriorated dyed textile samples simulated to ancient ones. *International Journal of Conservation Science*, 6(2).

⁴⁹ Prasad, R., 2018. Indigo—the crop that created history and then itself became history. *Indian J. Hist. Sci*, 53(3), pp.296-301.

⁵⁰ Elsayed, Y., 2025. Application of a Multi-Functional Acrylic Based Varnish Reinforced with Nanoparticles in the Preservation of Oil Paintings, *Journal of the Faculty of Archaeology, Cairo University*18(28), pp.375-395.

⁵¹ Yong-hua, R., Bian-xia, L. and Xiao-ning, S., 2012. Research on the Aging of Natural Fiber Textiles. *Advances in Biomedical Engineering*, 9, p.1.

moisture at 60°C. After weathering, samples were kept at 22°C for 72 hrs. before measurements^{52,53}.

2.9. Microbial strains and cultivation (Microbial ageing):

Many fungi and bacteria were identified in the case-study object, so this part of the study focuses on assessing the efficiency of the T-CMs and T-SMs against the most prevalent fungi and bacteria identified in the case-study object. *Aspergillus niger* (*A. niger*), and *Fusarium solani* (*F. solani*) were the top prevalent tested fungi. *Staphylococcus aureus* (*S. aureus*), *Streptomyces albus* (*S. albus*), and *Bacillus cereus* (*B. cereus*) were the top prevalent tested gram-positive bacteria. *Escherichia coli* (*E. coli*), *Pseudomonas aeruginosa* (*P. aeruginosa*), *Candida albicans* (*C. albicans*) were the top prevalent tested gram-negative bacteria, according to the microscopic characteristics and simple staining, and gram and spore staining.

(PDA) and (NA) were used as media for fungal growth and bacterial growth, respectively. 5 ml of 0.09% NaCl saline solution was added to each slant. The cells were gently loosened by brushing with a sterile inoculating loop. A vortex mixer was used for 1 min. to remove microbial cells. Petri dishes containing 20 ml of NA medium were used to determine the antimicrobial activity⁵⁴. Each T-CMs and T-SMs (1cm×1cm) was placed above the microbial media. All fungal and bacterial strains were grown on PDA and NA slants and incubated at 30°C and 37°C, respectively. Zones of inhibition (ZOI) were measured after 1 day and 4 days of bacterial and fungal incubation, respectively⁵⁵.

3. RESULTS AND DISCUSSION

3.1. Characterization of the case-study object

3.1.1. Fibers

The SEM obtained micrographs of investigated fibers revealed that both wraps (Fig. 2a) and weft fibers (Fig. 2b) are from cotton fibers, whereas the dyed fibers used in embroidery (Fig. 2c) are made from silk fibers. Moreover, the SEM obtained micrographs showed the huge amount of dust and soiling matter accumulate fibers, which suffer brittleness and weakness.

⁵² Vilaplana, F., Nilsson, J., Sommer, D.V. and Karlsson, S., 2015. Analytical markers for silk degradation: comparing historic silk and silk artificially aged in different environments. *Analytical and bioanalytical chemistry*, 407(5), pp.1433-1449.

⁵³ Yasuda, K., Okazaki, Y., Abe, Y. and Tsuga, K., 2017. Effective UV/Ozone irradiation method for decontamination of hydroxyapatite surfaces. *Heliyon*, 3(8).

⁵⁴ El-Kadi, S.M., Mahmoud, M.K., Sayed-Ahmed, K.A. and El-Hendawy, M.A., 2018. Comparison between silver nanoparticles and silver nitrate as antifungal agent. *International Journal of Nanoscience and Nanoengineering*, 4(1), pp.5-1.

⁵⁵ Elsayed, Y., El-Kadi, S., El-Rian, M. and Mabrouk, N., 2023. The efficiency of microbial culture extracts as green antimicrobial products against some microorganisms colonizing the historic oil paintings. *Scientific Culture*, 9(2), pp.127-143.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

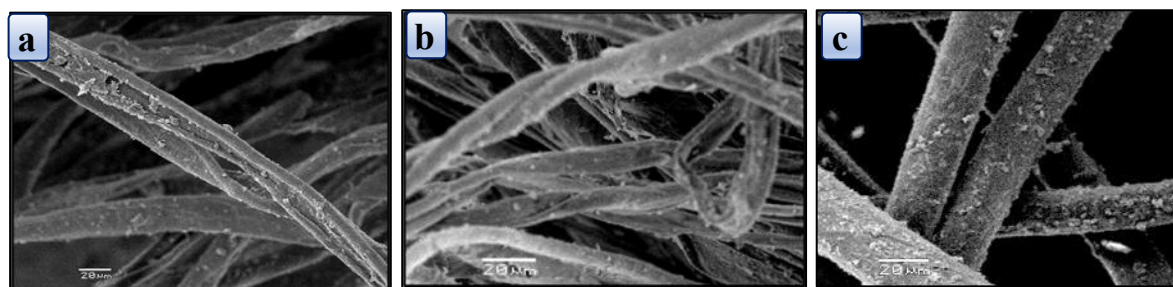


Fig. 2 SEM micrographs; wrap(a) and weft (b) of the base are cotton, and the embroidery(c) are silk.

3.1.2. Dyes and mordants

The results obtained from MALDI-ToF-MS and EDAX analysis of the seven dyed microsamples collected from the case-study object are shown in table 1. The results revealed that indigo was used in samples 1 and 2. Indigo also was used in sample 3 in addition to weld to get the greenish blue color. Cochineal and weld were used in samples 4 and 7 respectively. Madder was used in samples 5 and 6⁵⁶. Alum ($KAl(SO_4)_2 \cdot 12H_2O$) is the most suggested mordant in samples 3, 4, 5, and 7, due to the presence of the Al element. Ferrous sulfate ($FeSO_4$) appeared to be used in sample 6, maybe to transfer the red color of madder to brown. Samples 1 and 2 (dyed with indigo) are supposed to have no mordant, indigo is a substantive dye, so it adheres to fibers without the need for a chemical binding agent like a mordant, and the presence of the Al and Fe may be attributed to soiling matters. The presence of Fe in most samples, even in the blue ones, suggests that Fe isn't a mordant and it could be attributed to the soiling matters^{57,58}.

Table 1. Results of the MALDI-ToF-MS and EDAX in identifying dyes and mordants in the case-study object.

No.	Color	<i>m/z</i>	Detected compounds	Raw formula	Dye source	EDAX elements	Proposed mordant
1	Light blue	261	Indigotin	$C_{16}H_{10}N_2O_2$	Indigo, <i>Indigofera tinctoria</i>	C, N, O, Na, Mg, Al, Si, P, K, Ca, Fe	--
		261	Indirubin	$C_{16}H_{10}N_2O_2$			
2	Dark blue	261	Indigotin	$C_{16}H_{10}N_2O_2$	Indigo, <i>Indigofera tinctoria</i>	C, N, O, F, Na, Mg, Al, Si, S, Cl, K, Ca, Fe	--
		261	Indirubin	$C_{16}H_{10}N_2O_2$			
3	Bluish green	261	Indigotin	$C_{16}H_{10}N_2O_2$	Indigo, <i>Indigofera tinctoria</i> + Weld, <i>Reseda luteola</i>	C, N, O, Na, Mg, Al, Si, P, S, Cl, K, Sn, Ca, I, Fe	Alum
		262	Indirubin	$C_{16}H_{10}N_2O_2$			
		285	Luteolin	$C_{15}H_{10}O_6$			
4	Purple red	492	Carminic acid	$C_{22}H_{20}O_{13}$	Cochineal, <i>Dactylopius coccus</i>	C, N, O, Na, Mg, Al, Si, Cl, K, Ca, I	Alum

⁵⁶ Mabrouk, N., 2020. Archaeometrical study of a rare embroidered and appliqued leather tapestry from the safavid artworks. part i: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, 20(1), pp.163-163.

⁵⁷ Trojanowicz, et al, Chromatographic investigation, *Studies in conservation* (2004), 49 (2), 115-130.

⁵⁸ Mabrouk, N., 2020. Investigation and analysis of two Coptic textile fragments in the agricultural museum in Egypt. *Shedet*, 7(7), pp.277-297.

5	Light brown	239	Alizarin	C ₁₄ H ₈ O ₄	Madder, <i>Rubia tinctorum</i>	C, N, O, Mg, Al, Si, S, Cl, K, Ca, Fe	Alum
		255	Purpurin	C ₁₄ H ₈ O ₅			
6	Dark brown	239	Alizarin	C ₁₄ H ₈ O ₄	Madder, <i>Rubia tinctorum</i>	C, N, O, Na, Mg, Si, P, Cl, K, Ca, Fe	Ferrous sulfate
		255	Purpurin	C ₁₄ H ₈ O ₅			
7	Yellow	270	Apigenin	C ₁₅ H ₁₀ O ₅	Weld, <i>Reseda luteola</i>	C, N, O, F, Na, Mg, Al, Si, Cl, K, Sn, Ca	Alum
		286	Luteolin	C ₁₅ H ₁₀ O ₆			

3.2. Characterization of NC

The XRD spectrum obtained of the prepared Ag/ZnO-NC (Fig. 3a) has confirmed that the NC has been successfully prepared. It exhibits three peaks of Ag at 38.21°, 44.25°, 64.43°, and 78.12° which match with (111), (200), (220), and (311) planes respectively, and are indexed to the structure of Ag-NPs according to card No. 4-783, JCPDS^{59,60}. Seven peaks of ZnO were exhibited at 32.34°, 34.63°, 36.45°, 47.5°, 56.73°, 63.07°, and 67.95° which match with the (100), (002), (101), (102), (110), (103), and (112) planes, respectively, and are indexed to the structure of ZnO-NPs according to the card No. 36-1451, JCPDS⁶¹. The EDAX spectrum obtained of the prepared Ag/ZnO-NC (Fig. 3b) has also confirmed that the NC has been successfully prepared. (Zn), (Ag), and (O) signals were observed; these elements indicate the presence of ZnO loaded with Ag⁶². The spectral results obtained from XRD and EDAX were confirmed by TEM and SEM (Fig. 3c, d), which exhibited an average diameter of ~25 nm. It shows the successful synthesis of Ag/ZnO-NC, where a dispersion of the spherical Ag-NPs anchored on the surface of ZnO-NPs⁶³.

⁵⁹ Garibo, D., Borbón-Núñez, H.A., de León, J.N.D., García Mendoza, E., Estrada, I., Toledano-Magaña, Y., Tiznado, H., Ovalle-Marroquin, M., Soto-Ramos, A.G., Blanco, A. and Rodríguez, J.A., 2020. Green synthesis of silver nanoparticles using *Lysilomaacapulcensis* exhibit high-antimicrobial activity. *Scientific reports*, 10(1), p.12805.

⁶⁰ Yadav, L.R., Pratibha, S., Manjunath, K., Shivanna, M., Ramakrishnappa, T., Dhananjaya, N. and Nagaraju, G., 2019. Green synthesis of AgZnO nanoparticles: Structural analysis, hydrogen generation, formylation and biodiesel applications. *Journal of Science: Advanced Materials and Devices*, 4(3), pp.425-431.

⁶¹ Alharthi, F.A., Alghamdi, A.A., Al-Zaqri, N., Alanazi, H.S., Alsyahi, A.A., Marghany, A.E. and Ahmad, N., 2020. Facile one-pot green synthesis of Ag–ZnO Nanocomposites using potato peel and their Ag concentration dependent photocatalytic properties. *Scientific Reports*, 10(1), p.20229.

⁶² Mtavangu, S.G., Machunda, R.L., van der Bruggen, B. and Njau, K.N., 2022. In situ facile green synthesis of Ag–ZnO nanocomposites using *Tetradeniariperia* leaf extract and its antimicrobial efficacy on water disinfection. *Scientific Reports*, 12(1), p.15359.

⁶³ Mtavangu, S.G., Machunda, R.L., van der Bruggen, B. and Njau, K.N., 2022. In situ facile green synthesis of Ag–ZnO nanocomposites using *Tetradeniariperia* leaf extract and its antimicrobial efficacy on water disinfection. *Scientific Reports*, 12(1), p.15359.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

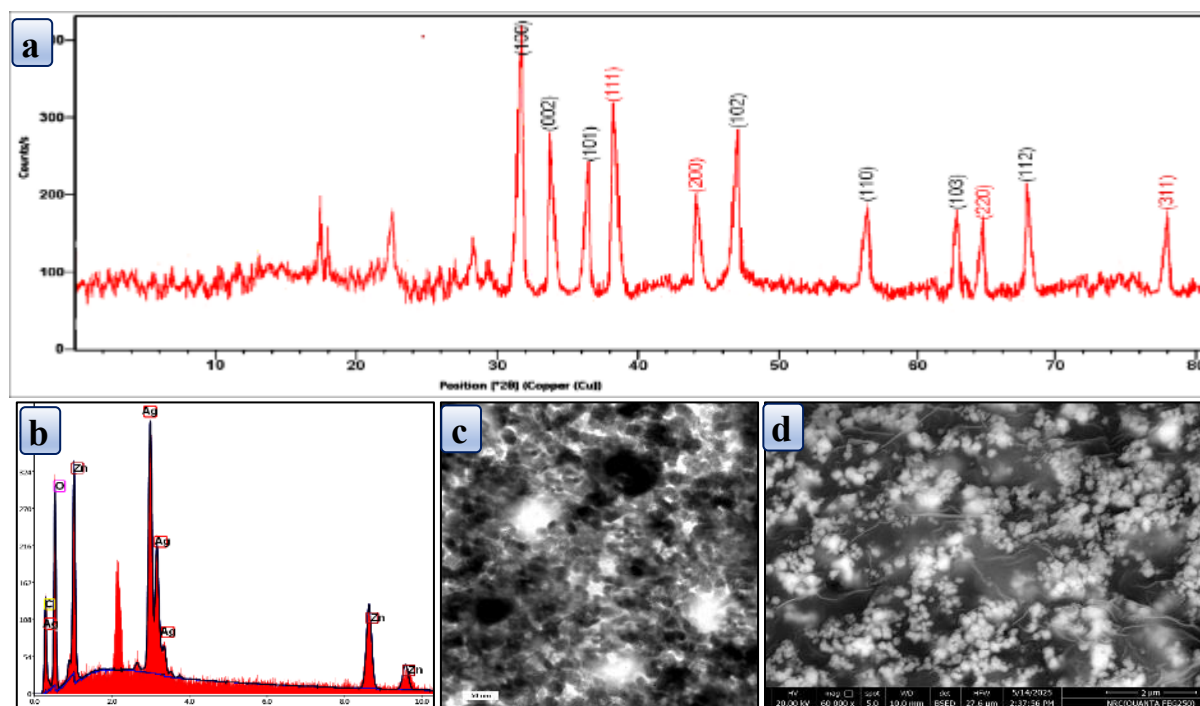


Fig. 3 Characterization results of the studied Ag/ZnO-NC, (a) XRD spectra, (b) EDAX spectra, (c) SEM micrograph, (d) TEM micrograph.

3.3. Characterization of fabric mock-ups

3.3.1. Mechanical properties

Tensile strength (n/cm^2) TS, elongation% (E%), and stiffness (BL) are the top important mechanical properties to be measured as the study targeted the consolidation of the tested fabric mock-ups. It is important to assess how far the proposed treatments strengthened the fabrics, and how far they preserved the flexibility of the fabrics as well. The results obtained from TS, E%, and BL of the untreated control cotton mock-ups (UT- CMs) were 345 n/cm^2 , 15%, and 30 mm respectively, and 265 n/cm^2 , 16%, and 35 mm in untreated control silk mock-ups (UT- SMs). The results obtained from TS (Fig. 4a), E% (Fig. 4b), and BL (Fig. 4c) of treated cotton mock-ups (T-CMs) and treated silk mock-ups (T-SMs) revealed that all treatments have increased all values of the TS, E% and BL of all samples in a variant ratio. Moreover, the results revealed that the more the addition of nanomaterials, the more the improvement of treatment properties and the more the increase of stiffness, which matches other literature results^{64,65}. Fig. 3 shows the differences (Δ) between treated and untreated mock-ups.

The increase of both TS and E% values in all treated samples is attributed to Beva371 acrylic copolymer, which caused more fiber cohesion. On the contrary, the values have decreased after aging, which is commonly due to the degradative effect of

⁶⁴MahovićPoljaček, S., Tomašegović, T., Leskovšek, M. and Stanković Elesini, U., 2021. Effect of SiO₂ and TiO₂ nanoparticles on the performance of UV visible fluorescent coatings. *Coatings*, 11(8), p.928.

⁶⁵Rehan, M., El-Naggar, M.E., Al-Enizi, A.M., Alothman, A.A., Nafady, A. and Abdelhameed, R.M., 2021. Development of silk fibers decorated with the in situ synthesized silver and gold nanoparticles: antimicrobial activity and creatinine adsorption capacity. *Journal of Industrial and Engineering Chemistry*, 97, pp.584-596.

UV, heat, and humidity on Beva371 and fabric as well^{66,67,68}. The results obtained from the TS, E%, and BL of all samples, concentration of 1000 µg/mL isn't recommended to be used, as it notably increased the BL of all mock-ups, especially the silk ones.

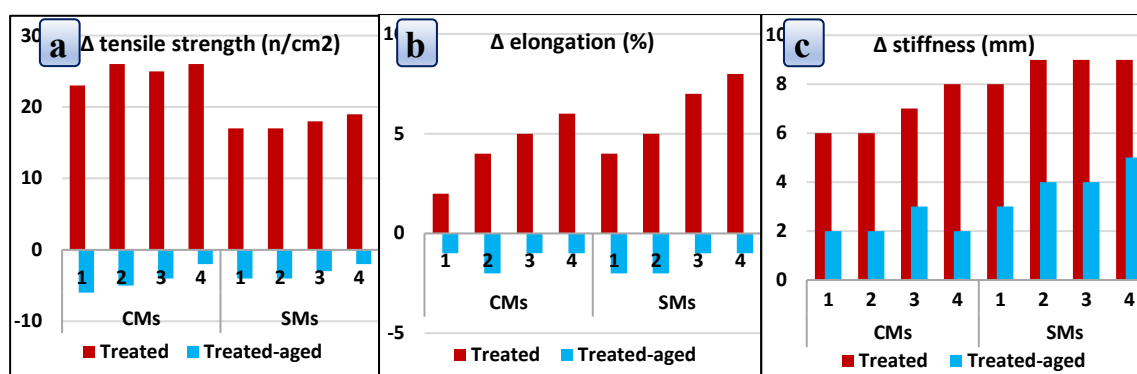


Fig. 4 Results of tensile strength n/cm² (a), elongation % (b), and stiffness mm (c) of treated, and treated-aged mock-ups.

3.3.2. Physical properties

3.3.2.1. Color differences

The results of both cotton and silk mock-ups after treatment (T-CMs and T-SMs) and after aging (TA-CMs and TA-SMs) (Fig. 5), compared to those of untreated mock-ups (UT-CMs and UT-SMs), revealed that Beva371 acrylic copolymer reinforced with Ag/ZnO-NC in the four tested concentrations (400, 600, 800, 1000 µg/mL) have variant effects on the color properties of mock-ups. The highest color difference (ΔE) of each sample after treatment and after aging is <3 (small difference), most ΔE is 0 to 1 (not detectable by eye), no remarkable differences (ΔE>3) were detected^{69,70}.

All T-CMs and T-SMs were differently affected by treatments, especially the concentration of 1000 µg/mL. T-SMs exhibited more ΔE, probably due to their dark colors of indigo dye, whilst, T-CMs exhibited less ΔE, probably due to their blank off-

⁶⁶ Al-Gaoudi, H.A., Marouf, M.A., Badry, N. and Rehan, M., 2024. A facile approach to developing multifunctional archaeological wool fabric surface with self-cleaning, and UV protection properties using nano-coating. *Inorganic Chemistry Communications*, 167, p.112716.

⁶⁷ Ezz Eldeen, H., Mohamed, W.S., Saad, H., Mahmoud, N. and Radi, R., 2020. A Comparative Study of the Impact of Beva 371 and Nano Vinyl Acetate Derivatives on Textiles Properties for Historical Textiles Conservation, Practical Application. *Egyptian Journal of Chemistry*, 63(4), pp.1383-1396.

⁶⁸ Ranakoti, L., Gangil, B., Rajesh, P.K., Singh, T., Sharma, S., Li, C., Ilyas, R.A. and Mahmoud, O., 2022. Effect of surface treatment and fiber loading on the physical, mechanical, sliding wear, and morphological characteristics of tasar silk fiber waste-epoxy composites for multifaceted biomedical and engineering applications: fabrication and characterizations. *journal of materials research and technology*, 19, pp.2863-2876.

⁶⁹ Marchiafava, V., Bartolozzi, G., Cucci, C., De Vita, M. and Picollo, M., 2014. Colour measurements for monitoring the conservation of contemporary artworks. *Journal of the International Colour Association*, 13, pp.36-42.

⁷⁰ Schanda, J. ed., 2007. *Colorimetry: understanding the CIE system*. John Wiley & Sons. pp. 79-91.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

white color, where, the main alteration in T-SMs caused by all treatment is the increase of ΔL^* , which is attributed to the white color of ZnO-NPs, but this white color wasn't detected in the white cotton fabric. After aging, the concentration of 1000 $\mu\text{g/mL}$ exhibited the most protective concentration against UV radiation, especially in T-SMs, which confirm the efficiency of the tested nanocomposite in protecting the dyed fabric from deteriorative agents, in accordance with literature^{71,72}.

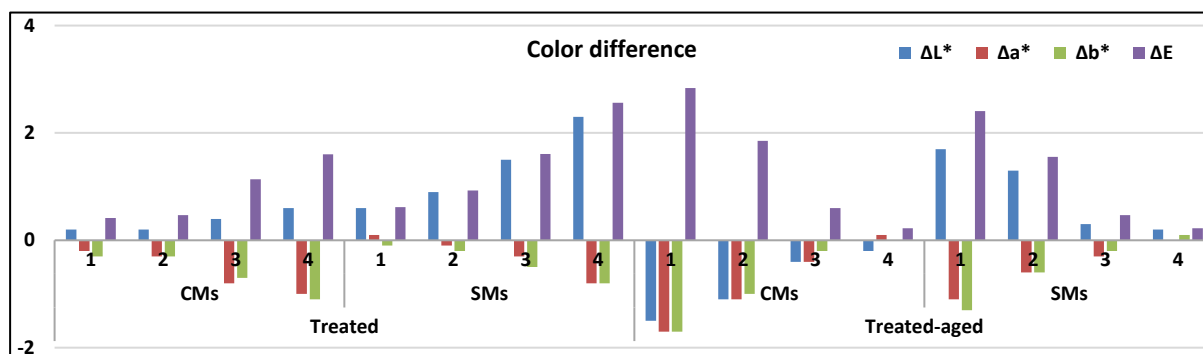


Fig. 5 Results of color difference of treated samples (a) and treated and UV-aged samples (b).

3.3.2.2. Air permeability and weight differences

The results obtained from air permeability and weight differences between CMs and SMs before, after treatment with the four concentrations, and after aging confirmed that no notable changes were revealed either in air permeability (Fig. 6a) or weight (Fig. 6b). Due to that undetectable effect on the properties of both fabrics, it could be concluded that the tested treatments are suitable proposed treatments for both cotton and silk fabrics as confirmed. The less air permeability of the fabric mock-ups is attributed to the filling of fabric micropores by Beva371 copolymer and nanomaterials; consequently, an increase in the weight of fabric samples^{73,74}.

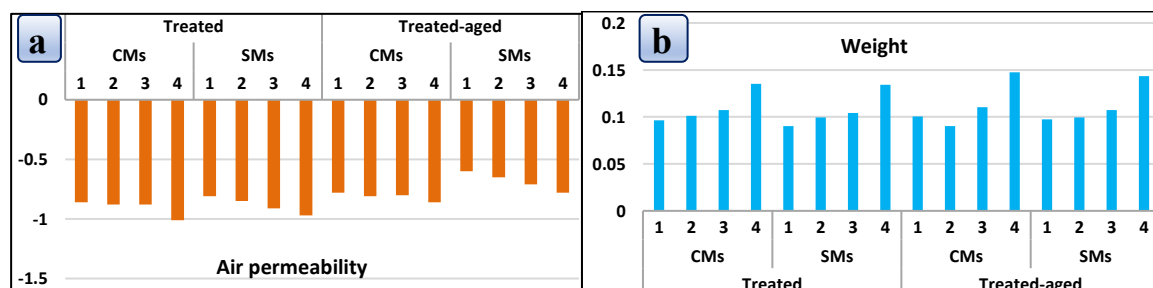


Fig. 6 results of air permeability (a), and weight (b) of treated, and treated-aged mock-ups

⁷¹ Abo-Elmaaref, M., Marouf, M., Mohamed, W.S. and Abdel Wahab, W.A., 2023. Antifungal and consolidation properties of linen textiles treated with silver nanoparticles loaded on hydroxypropyl cellulose polymer. *Heritage Science*, 11(1), p.120.

⁷² Ion, R.M. and Grigorescu, R.M., 2018. Polymeric Micro-and Nanosystems for Wood Artifacts. *New Uses of Micro and Nanomaterials*, p.73.

⁷³ Ahmed, H.E., Marouf, S. and Mohamed, W.S., 2022. Antifungal activity assessment of nanocomposites of natural chitosan and gelatin with a mahogany plant extract for conservation of historical textiles. *Heritage Science*, 10(1), p.198.

⁷⁴ Ogulata, R.T., 2006. Air permeability of woven fabrics. *Journal of Textile and Apparel, Technology and management*, 5(2), pp.1-10.

3.3.2.3. Antimicrobial activities

The results obtained from the antimicrobial activities of the three treatments (400, 600, and 800 µg/mL) for CMs and SMs against the above-mentioned eight microbes (table 2) revealed that all treatments protected the CMs and SMs against the tested microbes. The higher the concentration of Ag/ZnO-NC in Beva371, the higher the zone of inhibition (ZOI) in the Petri dish of the cultivated microbe⁷⁵. The antimicrobial activity of the Ag/ZnO-NC is mainly attributed to the effect of Ag, as it was concluded that Ag/ZnO-NC is a more efficient antimicrobial agent if compared with ZnO-NP⁷⁶. Moreover, the application of Ag/ZnO-NC usually exhibits better results than the application of individual Ag-NPs or ZnO-NPs⁷⁷. Regarding fungi, the treated mock-ups with a concentration of 800 µg/mL exhibited a higher ZOI against *A. niger* than *F. solani*; in complete accordance with literature⁷⁸, the ZOI was about 455 mm².

As for bacteria, the treated mock-ups with concentration of 800 µg/mL exhibited the highest ZOI in *E. coli*, the ZOI was about 525 mm², it is attributed to the cell wall of *E. coli*, which contains a thin peptidoglycan layer, unlike some other types of bacteria which contain a thick layer. This thin layer in *E. coli* helps facilitating the penetration of Ag/ZnO-NC⁷⁹. Other literature confirmed the same result^{80,81,82}. The *S. aureus* bacteria

⁷⁵ Jadhav, P., Shinde, S., Suryawanshi, S.S., Teli, B., Patil, P.S., Ramteke, A.A., Hiremath, N.G. and Prasad, N.R., 2020. Green AgNPs decorated ZnO nanocomposites for dye degradation and antimicrobial applications. *Engineered Science*, 12(21), pp.79-94.

⁷⁶ Zare, M., Namratha, K., Alghamdi, S., Mohammad, Y.H.E., Hezam, A., Zare, M., Drmosh, Q.A., Byrappa, K., Chandrashekar, B.N., Ramakrishna, S. and Zhang, X., 2019. Novel green biomimetic approach for synthesis of ZnO-Ag nanocomposite; antimicrobial activity against food-borne pathogen, biocompatibility and solar photocatalysis. *Scientific reports*, 9(1), p.8303.

⁷⁷ Dongale, T.D., Mullani, N.B., Patil, V.B., Tikke, R.S., Pawar, P.S., Mohite, S.V., Teli, A.M., Bagade, A.A., Pawar, K.K., Khot, K.V. and Shinde, S.S., 2018. Mimicking the biological synapse functions of analog memory, synaptic weights, and forgetting with ZnO-based memristive devices. *Journal of Nanoscience and Nanotechnology*, 18(11), pp.7758-7766.

⁷⁸ Jadhav, P., Shinde, S., Suryawanshi, S.S., Teli, B., Patil, P.S., Ramteke, A.A., Hiremath, N.G. and Prasad, N.R., 2020. Green AgNPs decorated ZnO nanocomposites for dye degradation and antimicrobial applications. *Engineered Science*, 12(21), pp.79-94.

⁷⁹ Ranjithkumar, B., Kumar, E.R., Srinivas, M., Ramalingam, H.B., Srinivas, C., Magesh, G., Balamurugan, A., Rahale, C.S. and ChandarShekar, B., 2021. Evaluation of structural, surface morphological and thermal properties of Ag-doped ZnO nanoparticles for antimicrobial activities. *Physica E: Low-dimensional Systems and Nanostructures*, 133, p.114801.

⁸⁰ Mtavangu, S.G., Machunda, R.L., van der Bruggen, B. and Njau, K.N., 2022. In situ facile green synthesis of Ag–ZnO nanocomposites using Tetradeniariperia leaf extract and its antimicrobial efficacy on water disinfection. *Scientific Reports*, 12(1), p.15359.

⁸¹ Azizi, S., Mohamad, R., Rahim, R.A., Moghaddam, A.B., Moniri, M., Ariff, A., Saad, W.Z. and Namvab, F., 2016. ZnO-Ag core shell nanocomposite formed by green method using essential oil of wild ginger and their bactericidal and cytotoxic effects. *Applied Surface Science*, 384, pp.517-524.

⁸² Ranjithkumar, B., Kumar, E.R., Srinivas, M., Ramalingam, H.B., Srinivas, C., Magesh, G., Balamurugan, A., Rahale, C.S. and ChandarShekar, B., 2021. Evaluation of structural,

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

exhibited some resistance to the Ag/ZnO-NC, probably due to its cell wall which is covered by a solid and thick peptidoglycan layer crosslinked by peptide chains, this layer protected the cell wall by hindering the penetration of Ag/ZnO-NC⁸³. The ZOI was about 220 mm², i.e. the Ag/ZnO-NC prevented the bacterial growth of the *S. aureus* upon the fabric mock-ups and little bit more.

Table 2. Antimicrobial activity by ZOI (mm²) of different treatments against the tested eight microbes.

Type	Name	T-CMs			T-SMs		
		400 µg/mL	600 µg/mL	800 µg/mL	400 µg/mL	600 µg/mL	800 µg/mL
Fungi	<i>A. niger</i>	215	339	455	192	307	414
	<i>F. solani</i>	184	276	365	161	244	324
Gram positive bacteria	<i>B. cereus</i>	187	280	325	164	248	284
	<i>S. albus</i>	174	211	299	151	179	258
	<i>S. aureus</i>	155	178	220	132	146	179
Gram negative bacteria	<i>E. coli</i>	317	402	525	294	370	484
	<i>P. aeruginosa</i>	231	290	413	208	258	372
	<i>C. albicans</i>	187	265	343	164	233	302

4. Treatment of the case-study object

4.1. Cleaning procedures

As mentioned above, the case-study object is suffering many deterioration phenomena, especially dust, stains, soiling matter, nails, etc. At the beginning, these deterioration features were treated by mechanical cleaning (Fig.7a, b) using small and soft brushes, cotton swabs, and scalpels. Soft blower was jointly used in vacuum cleaning and suctioning the eliminated matters to keep it away from the cleaned areas^{84,85,86}. In addition to the high acidity in the whole object, many focused spots exhibited very high acidity, probably due to the dyeing materials used, that's why Ca(OH)₂ was used in mitigating these high-acidity spots⁸⁷(Fig.7c).

surface morphological and thermal properties of Ag-doped ZnO nanoparticles for antimicrobial activities. *Physica E: Low-dimensional Systems and Nanostructures*, 133, p.114801.

⁸³ Nayak, D., Ashe, S., Rauta, P.R., Kumari, M. and Nayak, B., 2016. Bark extract mediated green synthesis of silver nanoparticles: evaluation of antimicrobial activity and antiproliferative response against osteosarcoma. *Materials Science and Engineering: C*, 58, pp.44-52.

⁸⁴ Landi S. (1992). *The Textile Conservation's Manual*, 2nd ed., Butterworth-Heinemann, Oxford.

⁸⁵ Marouf, M. and Saber, M., 2009. Treatment and Conservation of Archaeological Garment from Greco-Roman Period–The Egyptian Museum, Cairo. In *Proceedings, 4th International Congress on Science and Technology for the Safeguard of Cultural Heritage in the Mediterranean Basin*.

⁸⁶ Abdel-Kareem, O., Abdel-Rahman, A., Samaha, S.H. and Nasr, H.E., 2021. Conservation processes developed and adapted to clean, disinfect and reinforce a coptic textile object in Egypt. *Mediterranean Archaeology and Archaeometry*, 21(2), pp.181-181.

⁸⁷ Saoud, Khaled M., Imen Ibala, Dana El Ladki, Omar Ezzeldeen, and Shaukat Saeed. "Microwave assisted preparation of calcium hydroxide and barium hydroxide nanoparticles

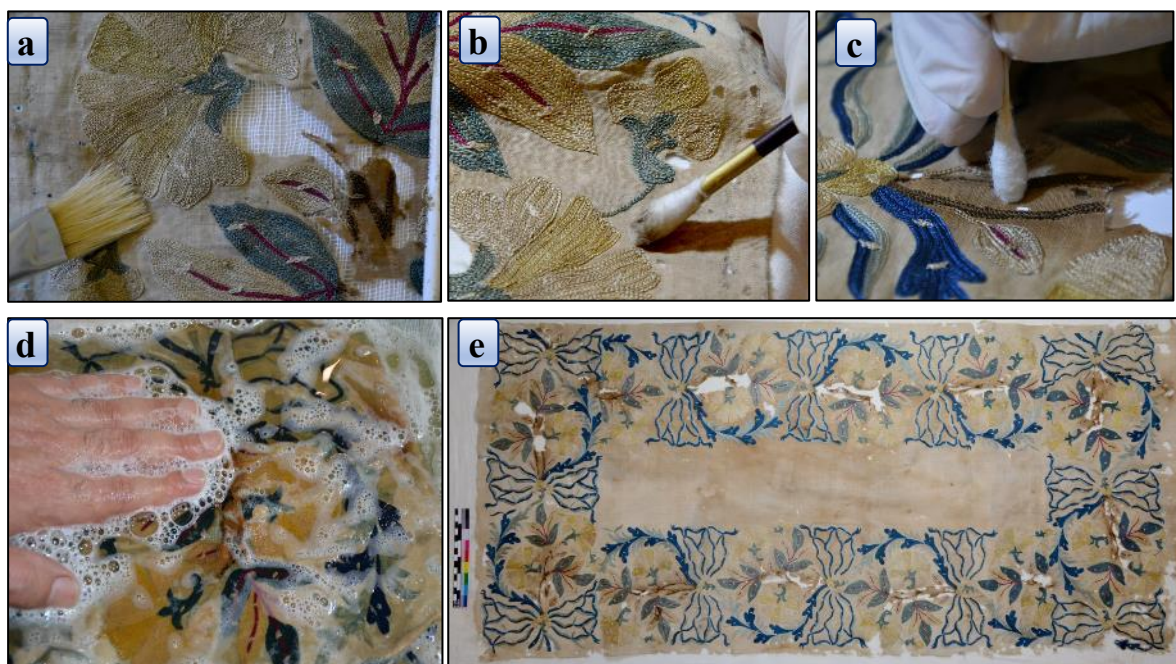


Fig. 7 treatment of the case-study object; (a, b) cleaning, (c) deacidification, (d) washing, (e) after drying.

The surface-accumulated and soiling matters were mechanically cleaned, as described above, but very fine matters of dust and air particulates were already still penetrating the fibers and the inner parts among threads. Washing the object was urgently needed to overcome this deteriorating phenomenon, in addition to neutralizing the high acidity in the whole object. A wooden frame covered with polyethylene was prepared in a suitable size and used as a washing bath. A water bath of 2% neutral soap in distilled water was prepared and poured into a washing bath. Distilled water in a hand sprayer was used to gradually humidify the object before immersing it in water. The object was sandwiched in tulle net and then put in the wash bath. To help swelling of all solid particulates and solve all water-soluble matters, the object was kept soaked in the bath for 20 minutes, then gently pressed by hand and sponge bulk to help penetrate water and remove soiling matters (Fig. 7d). The wash bath was then drained, and fresh distilled water was then put in, and this was repeated three times to rinse the object, and to get rid of all soiling matters and soap as well. The object was then lifted from the water bath and put upon a thick cotton towel, and covered by a similar one to absorb the water absorbed by the object fibers. The object was then fitted in the right direction, and all the crooked parts and threads were straightened. The object was then kept until dry at about 25°C^{88,89,90}(Fig.7e).

and their application for conservation of cultural heritage." In *Digital Heritage. Progress in Cultural Heritage: Documentation, Preservation, and Protection: 5th International Conference, EuroMed 2014, Limassol, Cyprus, November 3-8, 2014. Proceedings 5*, pp. 342-352. Springer International Publishing, 2014.

⁸⁸ Abdel-Kareem, O., Abdel-Rahman, A., Samaha, S.H. and Nasr, H.E., 2021. Conservation processes developed and adapted to clean, disinfect and reinforce a Coptic textile object in Egypt. *Mediterranean Archaeology and Archaeometry*, 21(2), pp.181-181.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

4.2. Application of Beva-based NC and consolidation

As mentioned above, the case-study object suffers brittleness and weakness, thus it urgently needs consolidation, either with needle works or with polymers or with both methods. For the consolidation of the case-study object, protecting it in the future from UV radiation, antimicrobial attacks, and humidity, our proposed Beva371-based Ag/ZnO-NC was applied. Despite the concentration of 800 $\mu\text{g/mL}$ in 2% Beva371 exhibited the top results, but it is suggested to apply concentration of 600 $\mu\text{g/mL}$, our study tested only the blue dyed fibers, other colors weren't tested. Moreover, the mock-ups aren't completely similar to the case-study object whatever we think that they are, so we found it is more suitable and safer to apply a lower concentration (600 $\mu\text{g/mL}$). A 50 ml handy sprayer filled with the Beva371-based Ag/ZnO-NC was used to typically spray both the recto and verso of the object and let it dry (Fig. 8a).

After the application of Beva-based NC, it was necessary to support the object with a new linen fabric, as follows^{91,92,93}: 1) Preparing a wooden frame coated with polyurethane varnish. 2) Washing a linen fabric in water and soap for 30 minutes at 100°C, then drying and ironing. 3) Stretching the washed linen fabric on the frame(Fig. 8b). 4) fixing the object to the support by fine needles and silk fibers dyed with suitable colors(Fig. 8c). Finally, the object was ready to be redisplayed in its showcase(Fig. 8d).



⁸⁹ Marouf, M. and Saber, M., 2009. Treatment and Conservation of Archaeological Garment from Greco-Roman Period–The Egyptian Museum, Cairo. In *Proceedings, 4th International Congress on Science and Technology for the Safeguard of Cultural Heritage in the Mediterranean Basin*.

⁹⁰ Matteini, M., Tosini, I., Giorgi, M. and Palei, G., 1999. Evaluation of possible methods of cleaning the Opus Anglicanum cope of Pope Pius II. In *Triennial meeting (12th), Lyon, 29 August-3 September 1999: preprints. Vol. 2* (pp. 625-630).

⁹¹ Landi S. (1992). *The Textile Conservation's Manual*, 2nd ed., Butterworth-Heinemann, Oxford.

⁹² Balazsy A. T. and Eastop D. (1998). *Chemical Principles of Textile Conservation*, Butterworth Heinemann, Oxford.

⁹³ Frances Lennard (2006). The art of tapestry conservation, in: Frances Lennard and Maria Hayward (eds), *Tapestry Conservation: Principles and Practice*, Butterworth-Heinemann, p.3-11.



Fig. 8 Consolidation and supporting of the case-study object; (a) application of Beva-based NC, (b) the linen support stretched on the frame, (c) fixing the object using needlework, (d) the object after final treatment.

5. CONCLUSIONS

The case-study object was classified as 15th-16th century AD Turkish tablecloth made of plain cotton base embroidered with silk threads, dyed with natural dyes of indigo, cochineal, weld, and madder, suffered many deterioration features like brittleness, weakness, high acidity, tears, ruptures, biodeterioration, dust and soiling matters. The experimental study implemented using blank cotton and indigo-dyed silk fabric mock-ups treated with Ag/ZnO NC showed the suitability of the 800 µg/mL Ag/ZnO NC suspended in 2% Beva371 in the treatment of mock-ups. The obtained results confirmed the improvement in the properties of treated mock-ups without notable side effects, especially in stiffness, color, and air permeability. After accelerated ageing, the treated mock-ups exhibited high resistance against UV, heat, moisture, fungi (especially *A. niger*), and bacteria (especially *E. coli*). The case-study object was then cleaned, deacidified, consolidated with the Beva371-Ag/ZnO NC, and supported with needlework on a new blank linen fabric stretched on a treated wooden frame.

ACKNOWLEDGEMENT

The author would like to thank Prof. El-Hussien Moawad at the Faculty of Science, Dr. Khaled Sayed-Ahmed, and Dr. Mohamed El-Rian at the Faculty of Agriculture, Damietta University, for preparation of nanomaterials, and antimicrobial experiments of the study.

Funding

No funding was received.

REFERENCES

- 1) Abdel-kareem, O., 2015. Preparation of experimental deteriorated dyed textile samples simulated to ancient ones. *International Journal of Conservation Science*, 6(2).
- 2) Abdel-Kareem, O., Abdel-Rahman, A., Samaha, S.H. and Nasr, H.E., 2021. Conservation processes developed and adapted to clean, disinfect and reinforce a Coptic textile object in Egypt. *Mediterranean Archaeology and Archaeometry*, 21(2), pp.181-181.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

- 3) Abo-Elmaaref, M., Marouf, M., Mohamed, W.S. and Abdel Wahab, W.A., 2023. Antifungal and consolidation properties of linen textiles treated with silver nanoparticles loaded on hydroxypropyl cellulose polymer. *Heritage Science*, 11(1), p.120.
- 4) Ahmed, H.E., Marouf, S. and Mohamed, W.S., 2022. Antifungal activity assessment of nanocomposites of natural chitosan and gelatin with a mahogany plant extract for conservation of historical textiles. *Heritage Science*, 10(1), p.198.
- 5) Akhter, M.S., Hossain, S.J., Hossain, S.A. and Datta, R.K., 2012. Isolation and characterization of salinity tolerant *Azotobacter* sp. *Greener J Biol Sci*, 2(3), pp.43-51.
- 6) Al-Gaoudi, H.A., Marouf, M.A., Badry, N. and Rehan, M., 2024. A facile approach to developing multifunctional archaeological wool fabric surface with self-cleaning, and UV protection properties using nano-coating. *Inorganic Chemistry Communications*, 167, p.112716.
- 7) Alharthi, F.A., Alghamdi, A.A., Al-Zaqri, N., Alanazi, H.S., Alsyahe, A.A., Marghany, A.E. and Ahmad, N., 2020. Facile one-pot green synthesis of Ag–ZnO Nanocomposites using potato peel and their Ag concentration dependent photocatalytic properties. *Scientific Reports*, 10(1), p.20229.
- 8) ASTM International, 2004. Annual book of ASTM standards. ASTM international.
- 9) Azizi, S., Mohamad, R., Rahim, R.A., Moghaddam, A.B., Moniri, M., Ariff, A., Saad, W.Z. and Namvab, F., 2016. ZnO-Ag core shell nanocomposite formed by green method using essential oil of wild ginger and their bactericidal and cytotoxic effects. *Applied Surface Science*, 384, pp.517-524.
- 10) Balazsy A. T. and Eastop D. (1998). *Chemical Principles of Textile Conservation*, Butterworth Heinemann, Oxford.
- 11) Becerra, J., Ortiz, P., Zaderenko, A.P. and Karapanagiotis, I., 2020. Assessment of nanoparticles/nanocomposites to inhibit micro-algal fouling on limestone façades. *Building Research & Information*, 48(2), pp.180-190.
- 12) Bechtold, T., Turcanu, A., Ganglberger, E. and Geissler, S., 2003. Natural dyes in modern textile dyehouses—how to combine experiences of two centuries to meet the demands of the future?. *Journal of Cleaner Production*, 11(5), pp.499-509.
- 13) Bhaduri, J., Kundu, P., Mitra, D. and Roy, S.K., 2016. Isolation and characterisation of Nitrogen Fixing Bacteria (*Azotobacter* sp.) from Tea field soil of Dooars and Darjeeling region of North Bengal, India. *Int. J. Eng. Sci. Invent*, 5, pp.46-51.
- 14) Bratasz, Ł., Łukomski, M., Klisińska-Kopacz, A., Zawadzki, W., Dzierżęga, K., Bartosik, M., Sobczyk, J., Lennard, F.J. and Kozłowski, R., 2015. Risk of Climate-Induced Damage in Historic Textiles. *Strain*, 51(1), pp.78-88.
- 15) Dhage, S. R., Renu Pasricha, and V. Ravi. "Synthesis of fine particles of ZnO at 100 C." *Materials Letters* 59, no. 7 (2005): 779-781.
- 16) Domsch, K.H., Gams, W. and Anderson, T.H., 1980. *Compendium of soil fungi*. Volume 1.
- 17) Dongale, T.D., Mullani, N.B., Patil, V.B., Tikke, R.S., Pawar, P.S., Mohite, S.V., Teli, A.M., Bagade, A.A., Pawar, K.K., Khot, K.V. and Shinde, S.S., 2018. Mimicking the biological synapse functions of analog memory, synaptic weights, and forgetting with ZnO-based memristive devices. *Journal of Nanoscience and Nanotechnology*, 18(11), pp.7758-7766.
- 18) El-Kadi, S.M., Mahmoud, M.K., Sayed-Ahmed, K.A. and El-Hendawy, M.A., 2018. Comparison between silver nanoparticles and silver nitrate as antifungal agent. *International Journal of Nanoscience and Nanoengineering*, 4(1), pp.5-1.

- 19) Elsayed, Y. and Sayed-Ahmed, K., 2024. A Novel Approach of Using Green Synthesized Tellurium Nanoparticles to Protect Historical Oil Paintings from Bacterial Degradation. *Shedet*, 13(13).
- 20) Elsayed, Y., 2025. Application of a Multi-Functional Acrylic Based Varnish Reinforced with Nanoparticles in the Preservation of Oil Paintings. *مجلة كلية الآثار. جامعة القاهرة*, 18(28), pp.375-395.
- 21) Elsayed, Y., El-Kadi, S., El-Rian, M. and Mabrouk, N., 2023. The efficiency of microbial culture extracts as green antimicrobial products against some microorganisms colonizing the historic oil paintings. *Scientific Culture*, 9(2), pp.127-143.
- 22) Ezz Eldeen, H., Mohamed, W.S., Saad, H., Mahmoud, N. and Radi, R., 2020. A Comparative Study of the Impact of Beva 371 and Nano Vinyl Acetate Derivatives on Textiles Properties for Historical Textiles Conservation, Practical Application. *Egyptian Journal of Chemistry*, 63(4), pp.1383-1396.
- 23) Frances Lennard (2006). The art of tapestry conservation, in: Frances Lennard and Maria Hayward (eds), *Tapestry Conservation: Principles and Practice*, Butterworth-Heinemann, p.3-11.
- 24) Garibo, D., Borbón-Núñez, H.A., de León, J.N.D., García Mendoza, E., Estrada, I., Toledano-Magaña, Y., Tiznado, H., Ovalle-Marroquin, M., Soto-Ramos, A.G., Blanco, A. and Rodríguez, J.A., 2020. Green synthesis of silver nanoparticles using *Lysilomaacapulcensis* exhibit high-antimicrobial activity. *Scientific reports*, 10(1), p.12805.
- 25) Guo, C., Lan, L., Liu, Y., Meng, N. and Li, C., 2023. Comparison of environmental criteria for conservation and storage of collections: A comprehensive literature review. *Building and Environment*, 243, p.110665.
- 26) Ibrahim, M.M., Mohamed, W.S. and Mohamed, H.M., 2021. Experimental study for evaluation of Paraloid® B72 and its nanocomposite with nano TiO₂ and nano ZnO for consolidation of pottery samples. *Scientific culture*, 7(2), pp.101-111.
- 27) Ilieș, D.C., Hodor, N., Indrie, L., Dejeu, P., Ilieș, A., Albu, A., Caciara, T., Ilieș, M., Barbu-Tudoran, L. and Grama, V., 2021. Investigations of the surface of heritage objects and green bioremediation: case study of artefacts from Maramureș, Romania. *Applied Sciences*, 11(14), p.6643.
- 28) Ilies, D.C., Zlatev, Z., Ilies, A., Zharas, B., Pantea, E., Hodor, N., Indrie, L., Turza, A., Taghiyari, H.R., Caciara, T. and Costea, M., 2022. Interdisciplinary research to advance digital imagery and natural compounds for eco-cleaning and for preserving textile cultural heritage. *Sensors*, 22(12), p.4442.
- 29) Ion, R.M. and Grigorescu, R.M., 2018. Polymeric Micro-and Nanosystems for Wood Artifacts. *New Uses of Micro and Nanomaterials*, p.73.
- 30) Jadhav, P., Shinde, S., Suryawanshi, S.S., Teli, B., Patil, P.S., Ramteke, A.A., Hiremath, N.G. and Prasad, N.R., 2020. Green AgNPs decorated ZnO nanocomposites for dye degradation and antimicrobial applications. *Engineered Science*, 12(21), pp.79-94.
- 31) Jakiela, S., Bratasz, Ł. and Kozłowski, R., 2008. Numerical modelling of moisture movement and related stress field in lime wood subjected to changing climate conditions. *Wood Science and Technology*, 42, pp.21-37.
- 32) Kakakhel, M.A., Wu, F., Gu, J.D., Feng, H., Shah, K. and Wang, W., 2019. Controlling biodeterioration of cultural heritage objects with biocides: A review. *International Biodeterioration & Biodegradation*, 143, p.104721.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

- 33) Landi S. (1992). *The Textile Conservation's Manual*, 2nd ed., Butterworth-Heinemann, Oxford.
- 34) Mabrouk, N., 2020. Archaeometrical study of a rare embroidered and appliqued leather tapestry from the safavid artworks. part i: weaving fibers and dyes. *Mediterranean Archaeology and Archaeometry*, 20(1), pp.163-163.
- 35) Mabrouk, N., 2020. Investigation and analysis of two Coptic textile fragments in the agricultural museum in Egypt. *Shedet*, 7(7), pp.277-297.
- 36) Maekawa, S., Beltran, V.L. and Henry, M.C., 2015. *Environmental Management for Collections: Alternative Conservation Strategies for Hot and Humid Climates* (). Getty Publications.
- 37) MahovićPoljaček, S., Tomašegović, T., Leskovšek, M. and Stanković Elesini, U., 2021. Effect of SiO₂ and TiO₂ nanoparticles on the performance of UV visible fluorescent coatings. *Coatings*, 11(8), p.928.
- 38) Mandal, S., Kumar, P., Satpathy, B., Das, K. and Das, S., 2024. Nanostructured metal oxide based coating for the protection and conservation of cultural heritage: A comprehensive review. *Journal of Cultural Heritage*, 69, pp.94-112.
- 39) Marchiafava, V., Bartolozzi, G., Cucci, C., De Vita, M. and Picollo, M., 2014. Colour measurements for monitoring the conservation of contemporary artworks. *Journal of the International Colour Association*, 13, pp.36-42.
- 40) Marouf, M. and Saber, M., 2009. Treatment and Conservation of Archaeological Garment from Greco-Roman Period–The Egyptian Museum, Cairo. In *Proceedings, 4th International Congress on Science and Technology for the Safeguard of Cultural Heritage in the Mediterranean Basin*.
- 41) Matteini, M., Tosini, I., Giorgi, M. and Palei, G., 1999. Evaluation of possible methods of cleaning the Opus Anglicanum cope of Pope Pius II. In *Triennial meeting (12th)*, Lyon, 29 August-3 September 1999: preprints. Vol. 2 (pp. 625-630).
- 42) Michalski, S., 2007. The ideal climate, risk management, the ASHRAE chapter, proofed fluctuations, and towards a full risk analysis model. *Experts roundtable on sustainable climate management strategies*, pp.1-19.
- 43) Mtavangu, S.G., Machunda, R.L., van der Bruggen, B. and Njau, K.N., 2022. In situ facile green synthesis of Ag–ZnO nanocomposites using *Tetradeniariperia* leaf extract and its antimicrobial efficacy on water disinfection. *Scientific Reports*, 12(1), p.15359.
- 44) Nayak, D., Ashe, S., Rauta, P.R., Kumari, M. and Nayak, B., 2016. Bark extract mediated green synthesis of silver nanoparticles: evaluation of antimicrobial activity and antiproliferative response against osteosarcoma. *Materials Science and Engineering: C*, 58, pp.44-52.
- 45) Noohpisheh, Z., Amiri, H., Farhadi, S. and Mohammadi-Gholami, A., 2020. Green synthesis of Ag-ZnO nanocomposites using *Trigonella foenum-graecum* leaf extract and their antibacterial, antifungal, antioxidant and photocatalytic properties. *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy*, 240, p.118595.
- 46) Ogulata, R.T., 2006. Air permeability of woven fabrics. *Journal of Textile and Apparel, Technology and management*, 5(2), pp.1-10.
- 47) Oves, M., Arshad, M., Khan, M.S., Ahmed, A.S., Azam, A. and Ismail, I.M., 2015. Anti-microbial activity of cobalt doped zinc oxide nanoparticles: Targeting water borne bacteria. *Journal of Saudi Chemical Society*, 19(5), pp.581-588.
- 48) Pekhtasheva, E.L., Neverov, A.N., Kubica, S. and Zaikov, G.E., 2011. Biodegradation and biodeterioration of some natural polymers. *Polymers Research Journal*, 5(1), p.77.

- 49) Perkas, N., G. Amirian, S. Dubinsky, S. Gazit, and A. Gedanken. "Ultrasound-assisted coating of nylon 6, 6 with silver nanoparticles and its antibacterial activity." *Journal of applied polymer science* 104, no. 3 (2007): 1423-1430.
- 50) Pietrzak, K., Puchalski, M., Otlewska, A., Wrzosek, H., Guiamet, P., Piotrowska, M. and Gutarowska, B., 2017. Microbial diversity of pre-Columbian archaeological textiles and the effect of silver nanoparticles misting disinfection. *Journal of Cultural Heritage*, 23, pp.138-147.
- 51) Podporska-Carroll, J., Myles, A., Quilty, B., McCormack, D.E., Fagan, R., Hinder, S.J., Dionysiou, D.D. and Pillai, S.C., 2017. Antibacterial properties of F-doped ZnO visible light photocatalyst. *Journal of hazardous materials*, 324, pp.39-47.
- 52) Prasad, R., 2018. Indigo—the crop that created history and then itself became history. *Indian J. Hist. Sci*, 53(3), pp.296-301.
- 53) Ranakoti, L., Gangil, B., Rajesh, P.K., Singh, T., Sharma, S., Li, C., Ilyas, R.A. and Mahmoud, O., 2022. Effect of surface treatment and fiber loading on the physical, mechanical, sliding wear, and morphological characteristics of tasar silk fiber waste-epoxy composites for multifaceted biomedical and engineering applications: fabrication and characterizations. *journal of materials research and technology*, 19, pp.2863-2876.
- 54) Ranjithkumar, B., Kumar, E.R., Srinivas, M., Ramalingam, H.B., Srinivas, C., Magesh, G., Balamurugan, A., Rahale, C.S. and ChandarShekar, B., 2021. Evaluation of structural, surface morphological and thermal properties of Ag-doped ZnO nanoparticles for antimicrobial activities. *Physica E: Low-dimensional Systems and Nanostructures*, 133, p.114801.
- 55) Rehan, M., El-Naggar, M.E., Al-Enizi, A.M., Alothman, A.A., Nafady, A. and Abdelhameed, R.M., 2021. Development of silk fibers decorated with the in situ synthesized silver and gold nanoparticles: antimicrobial activity and creatinine adsorption capacity. *Journal of Industrial and Engineering Chemistry*, 97, pp.584-596.
- 56) Romero-Pastor, J., Navas, N., Rodriguez-Simon, L., Lario-Simón, A., Kuckova, S. and Manzano, E., Application of Matrix-Assisted Laser Desorption and Ionization Time of Flight Mass Spectrometry to the Study of the Proteinaceous Binders in Paint, "The Life of Virgin" by Alonso Cano (17th Century) as a Case Study. *European Journal of Mass Spectrometry*, (2015), 21(1), pp.19-26.
- 57) Ruparelia, J.P., Chatterjee, A.K., Duttagupta, S.P. and Mukherji, S., 2008. Strain specificity in antimicrobial activity of silver and copper nanoparticles. *Acta biomaterialia*, 4(3), pp.707-716.
- 58) Saez, N.O., Vanden Berghe, I., Schalm, O., De Munck, B. and Caen, J., 2019. Material analysis versus historical dye recipes: ingredients found in black dyed wool from five Belgian archives (1650-1850). *Conservar Património*, 31, pp.115-132.
- 59) Samadelli, M., Zink, A.R., Roselli, G., Gabrielli, S., Tabandeh, S. and Fernicola, V.C., 2019. Development of passive controlled atmosphere display cases for the conservation of cultural assets. *Journal of Cultural Heritage*, 35, pp.145-153.
- 60) Saoud, Khaled M., Imen Ibala, Dana El Ladki, Omar Ezzeldeen, and Shaukat Saeed. "Microwave assisted preparation of calcium hydroxide and barium hydroxide nanoparticles and their application for conservation of cultural heritage." In *Digital Heritage. Progress in Cultural Heritage: Documentation, Preservation, and Protection: 5th International Conference, EuroMed 2014, Limassol, Cyprus, November 3-8, 2014. Proceedings 5*, pp. 342-352. Springer International Publishing, 2014.
- 61) Schanda, J. ed., 2007. *Colorimetry: understanding the CIE system*. John Wiley & Sons. pp. 79-91.

Assessing the Efficacy of Beva371-Suspended Ag/ZnO Nanocomposite for the Conservation of Heritage Textiles: An Experimental and Applied Study

- 62) Sengupta, S., Debnath, S. and Sengupta, A., 2016. Fabric bending behaviour testing instrument for technical textiles. *Measurement*, 87, pp.205-215.
- 63) Spielman-Sun, E., Zaikova, T., Dankovich, T., Yun, J., Ryan, M., Hutchison, J.E. and Lowry, G.V., 2018. Effect of silver concentration and chemical transformations on release and antibacterial efficacy in silver-containing textiles. *NanoImpact*, 11, pp.51-57.
- 64) Summerell, B.A., Salleh, B. and Leslie, J.F., 2003. A utilitarian approach to *Fusarium* identification. *Plant disease*, 87(2), pp.117-128.
- 65) Tímár-Balázsy, Á. and Eastop, D., 2012. *Chemical principles of textile conservation*. Routledge.
- 66) Tiwari, V., Mishra, N., Gadani, K., Solanki, P.S., Shah, N.A. and Tiwari, M., 2018. Mechanism of anti-bacterial activity of zinc oxide nanoparticle against carbapenem-resistant *Acinetobacter baumannii*. *Frontiers in microbiology*, 9, p.1218.
- 67) Trandafilović, L.V., Božanić, D.K., Dimitrijević-Branković, S., Luyt, A.S. and Djoković, V., 2012. Fabrication and antibacterial properties of ZnO–alginate nanocomposites. *Carbohydrate Polymers*, 88(1), pp.263-269.
- 68) Trojanowicz, et al, Chromatographic investigation, *Studies in conservation* (2004), 49 (2), 115-130.
- 69) Varaprasad, K., Raghavendra, G.M., Jayaramudu, T. and Seo, J., 2016. Nano zinc oxide–sodium alginate antibacterial cellulose fibres. *Carbohydrate polymers*, 135, pp.349-355.
- 70) Vilaplana, F., Nilsson, J., Sommer, D.V. and Karlsson, S., 2015. Analytical markers for silk degradation: comparing historic silk and silk artificially aged in different environments. *Analytical and bioanalytical chemistry*, 407(5), pp.1433-1449.
- 71) Wendt, J.A., Indrie, L., Dejeu, P., Albu, A., Ilieș, D.C., Costea, M., Caciora, T., Ilieș, G., Hodor, N., Josan, I. and Berdenov, Z., 2021. Natural sources in preventive conservation of naturally aged textiles.
- 72) Yadav, L.R., Pratibha, S., Manjunath, K., Shivanna, M., Ramakrishnappa, T., Dhananjaya, N. and Nagaraju, G., 2019. Green synthesis of AgZnO nanoparticles: Structural analysis, hydrogen generation, formylation and biodiesel applications. *Journal of Science: Advanced Materials and Devices*, 4(3), pp.425-431.
- 73) Yasuda, K., Okazaki, Y., Abe, Y. and Tsuga, K., 2017. Effective UV/Ozone irradiation method for decontamination of hydroxyapatite surfaces. *Heliyon*, 3(8).
- 74) Yong-hua, R., Bian-xia, L. and Xiao-ning, S., 2012. Research on the Aging of Natural Fiber Textiles. *Advances in Biomedical Engineering*, 9, p.1.
- 75) Zare, M., Namratha, K., Alghamdi, S., Mohammad, Y.H.E., Hezam, A., Zare, M., Drmosh, Q.A., Byrappa, K., Chandrashekar, B.N., Ramakrishna, S. and Zhang, X., 2019. Novel green biomimetic approach for synthesis of ZnO-Ag nanocomposite; antimicrobial activity against food-borne pathogen, biocompatibility and solar photocatalysis. *Scientific reports*, 9(1), p.8303.